

Western Mathematics

TOPIC REVISION QUESTIONS

Mathematics Standard 2

MS-M2 Working with Time

Section I – Multiple Choice Questions (1 Mark each)

Section II – Longer Questions (Marks indicated beside questions)

Section I**12 marks**

1. Mr Kelly trains the Mansfield football team and he sends them for 20 sprints at the start of each training session.

If each sprint takes 14 seconds, and they take 30 seconds to jog back from each, how long is the sprint session?

- (A) 12 minutes and 40 seconds
- (B) 14 minutes and 16 seconds
- (C) 14 minutes and 40 seconds
- (D) 36 minutes and 40 seconds

2. Hazel arrives home one afternoon at the time shown on her watch.

She went out again at 7:24 pm.

How long was she at home?



- (A) 3 hours 11 min
- (B) 3 hours 49 min
- (C) 4 hours 11 min
- (D) 4 hours 49 min

3. Jasmine is going to bed at the time shown on her phone and wants to set an alarm to wake her.

She wants to get seven and a half hours sleep and allow an extra quarter of an hour to get to sleep.

What time should she set the alarm for the next morning?



- (A) 5:15 am
- (B) 5:30 am
- (C) 5:45 am
- (D) 6:00 am

Questions 4 – 6 refer to the section of a railway timetable for a Monday afternoon which is shown below:

Station	Arrival and Departure				
Penrith	1:33	2:03	2:33	3:44	4:36
Kingswood	1:40	2:06	2:38	----	----
St Marys	----	2:09	----	----	----
Mount Druitt	----	2:15	----	----	----
Blacktown	1:48	2:22	2:48	3:59	4:51
Seven Hills	----	2:25	----	----	----
Westmead	1:55	2:28	2:55	----	4:58
Parramatta	1:58	2:31	2:58	4:10	5:01
Granville	----	2:35	----	4:14	5:04
Lidcombe	----	2:39	----	----	----
Strathfield	2:10	2:43	3:10	4:25	5:15
Redfern	----	2:50	----	----	----
Central	2:23	2:57	3:23	4:37	5:28

4. How much longer does the train that leaves Penrith at 2:03 take to reach Central compared to the 2:33 train?
- (A) 3 minutes
- (B) 7 minutes
- (C) 13 minutes
- (D) 17 minutes
5. Lucia wants to catch a train from Blacktown to Strathfield, so she can change to another train which leaves there at 3:04. She needs to allow 6 minutes to get from one platform to the other. What is the latest train that she can catch from Blacktown?
- (A) 1:48
- (B) 2:22
- (C) 2:48
- (D) 3:59
6. Dee leaves a meeting at 1:52 and has a 10-minute walk to Kingswood station. What is the earliest time that she can get home by train if she lives 5 minutes walk from Granville station?
- (A) 2:40
- (B) 4:14
- (C) 4:19
- (D) 5:09

7. Josie messages Terri and asks to meet at 19:50 on Thursday.
When Terri receives the message, it is 10:15 on Wednesday night.
How long is it from when she got the message to when they meet?
- (A) 20 hours and 5 minutes
(B) 20 hours and 35 minutes
(C) 21 hours and 5 minutes
(D) 21 hours and 35 minutes
8. Ngaire lives in Sydney (UTC+10) and rings Owen in Perth (UTC+8), in summer when there is daylight saving in Sydney but not in Perth.
If she makes her call at 7:00 pm, at what local time does Owen take the call?
- (A) 4:00 pm
(B) 5:00 pm
(C) 9:00 pm
(D) 10:00pm
9. Two ships, the Braveheart and the Hollandia are at sea.
The coordinates of the Braveheart are 45°W , 25°N and of the Hollandia are 15°W 10°N .
What is their time difference?
- (A) Hollandia is 1 hour ahead of Braveheart
(B) Hollandia is 1 hours behind Braveheart
(C) Hollandia is 2 hours ahead of Braveheart
(D) Hollandia is 2 hours behind Braveheart
10. Michael is in Moscow (UTC+3) and Jo is in Seattle (UTC−8).
When the time in Seattle is 6 pm on Thursday, what is the time in Moscow?
- (A) 7 am Thursday
(B) 1 pm Thursday
(C) 11 pm Thursday
(D) 5 am Friday

11. Anchorage in Alaska is UTC−10 and Rio de Janeiro in Brazil is UTC−3.

When the time in Anchorage is 10 pm on 1st July, what is the time in Rio de Janeiro?

- (A) 5 am on 30th June
- (B) 9 am on 1st July
- (C) 5 am on 2nd July
- (D) 11 am on 2nd July

12. Max is in Sydney (UTC+10) and messages Tilly in Los Angeles (UTC−8).

The message is sent at 9:00 pm on 7th August when there is daylight saving in Los Angeles, but not in Sydney.

What is the local time when Tilly receives the message, assuming it is instantaneous?

- (A) 3:00 am on 6th August
- (B) 3:00 am on 7th August
- (C) 4:00 am on 7th August
- (D) 8:00 pm on 7th August

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TOPIC REVISION QUESTIONS - Section II

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 13 (3 marks)**Marks**

- (a) A T20 cricket match starts at 7:40pm and finishes at 10:27 pm.
What is the duration of the match?

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- (b) A streamed movie has a playing time of 2 hours and 28 minutes and Souxie starts watching the movie at 19:43.
At what time will the movie end, if she pauses it for 18 minutes in the middle to get a snack?

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Question 14 (2 marks)**2**

Yania works four shifts in a week.

The total time of the four shifts is 32 hours.

The times for her first three shifts were

Monday : Seven hours and fifteen minutes

Wednesday : Eight hours and thirty five minutes

Thursday : Seven hours and forty six minutes

What was the time for her last shift on Friday?

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Question 15 (5 marks)

The timetable below is for a tramline on weekday mornings.

Station	Time				
Cromwell St	6:33	7:03	7:33	7:44	8:06
Wyatt	6:38	7:08	7:38	7:49	8:11
Tudor Towers	6:45	7:15		7:56	
Weston	6:51	7:21	7:50	8:02	8:23
Harris St	6:54	7:24		8:05	
Bryan Road	6:59	7:29	7:57	8:10	8:30
Woolsey	7:07	7:37	8:04	8:18	
Brereton Arr	7:16	7:46	8:13	8:27	8:46
Dep	7:20			8:32	
Poole	7:25			8:37	
Rochford Rail	7:31		8:21	8:43	
Richmond	7:35		8:25	8:47	
Seymour	7:41		8:31	8:53	
City Centre	7:46		8:36	8:58	

- (a) Wendell arrives at Cromwell St station at 7:20. What is the earliest he can get to City Centre? 1

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- (b) Whitney needs to catch a train from Rochford Rail at 8:45 and needs twelve minutes to get from the tram to the train. 2
What is the latest she can catch a tram from Woolsey Station?

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- (c) Jami used to live near the Weston station and caught the tram at 7:50 to get to work near the Seymour station. 2
She moved to a unit near the Harris St station.
What tram(s) does she have to catch so that she arrives at Seymour no later than she previously did?

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Question 16 (5 marks)

There are three members of an international forum, who live around the globe and chat online.

Jenna lives in Perth (UTC+8).

Sofia lives in Lisbon (UTC)

Brittney lives in Calgary (UTC−7).

Pablo lives in Buenos Aires (UTC−4)

- (a) What is the greatest time difference between the members?

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- (b) If Sofia initiates a chat between all members at 8:00 pm her local time, who is notified to join earliest in the morning?

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- (c) If Jenna starts a chat at 2:00 pm on Wed 21st July, what is the time and date that Brittney is notified to join?

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Question 17 (4 marks)

Tayla is on Utopia Island, (0°S , 165°E) and her brother is on Fantasy Island (0° , 150°W).
The islands are 4 500 km apart.

- (a) What is the time difference between the two islands? 1

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- (b) At 6:30 am on Saturday August 15th, she rings her brother on Fantasy Island. 1
What is the local time when her brother takes the call?

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- (c) Tayla flew to Fantasy Island in a plane which travelled at 250 km/h. 2
She left Utopia Island at 2:00 p.m. on Tuesday August 18th (Local time).
What was the local time on Fantasy Island when she arrived?

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Question 18 (3 marks).

Kerrie is in Nairobi, (UTC+3) and Mike is in Washington, (UTC-5).

- (a) Kerrie is going to ring Mike at 6 am on Tuesday, Nairobi time.
What day and time will it be in Washington when she rings?

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- (b) Mike is going to fly from Washington to Nairobi. His flight will leave on Wednesday
at 3 pm, Washington time, and will take 10 hours.

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What day and time will it be in Nairobi when he arrives?

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End of Revision

Western Mathematics MS-M2 Working with Time
Section I - Worked Solutions

Revision

No	Working	Answer
1	Each sprint = $14 + 30 = 44$ sec Time for session = $44 \text{ sec} \times 20 = 880 \text{ sec}$ Time in min = $\frac{880}{60} = 14.66666667 \text{ min}$ = 14 min and 40 sec	C
2	Time shown is 2: 35 p. m. or 14: 35 Find time from 2: 35 to 7: 24 Time = 25 min to 3: 00 + 4 hours to 7: 00 + 24 min to 7: 24 Time = 4 hours and 49 min	D
3	Time shown is 21: 45 Want time 7: 45 after this Time = $21:45 + 0:45 = 22:30 + 2:00 = 00:30 + 5:00 = 5:30 \text{ am}$	B
4	The 2: 03 train arrives at 2: 57 so takes $57 - 3 = 53 \text{ min}$ The 2: 33 train arrives at 3: 23 so takes $(60 - 33) + 23 = 50 \text{ min}$ Difference = $53 - 50 = 3 \text{ minutes}$	A
5	She needs to arrive 6 min before 3: 04 so at 2: 58 at the latest The latest arrival at Strathfield before 2: 58 is 2: 43 She needs to catch this train at Blacktown at 2: 22	B
6	She would to arrive at Kingswood station at $1:52 + 10 \text{ min} = 2:02$ The next train is at 2: 06 and arrives at Granville at 2: 35 After a 5 minute walk she would get home at 2: 40	A
7	Time from 10: 15 pm to midnight Wed is 1: 45 Time from 10: 15 pm Wed to 19: 50 Thur = $1:45 + 19:50$ = 20 hours + 95 min " = 21 hours + 35 min	D
8	7:00 pm daylight saving time is 6:00 pm standard time Sydney is + 2 hours UTC so in Perth it is 2 hours before 6 pm so it is 4 pm in Perth	A
9	Longitude difference gives time difference Longitude difference (Subtract since both $W = 45^\circ - 15^\circ = 30^\circ$ 360° longitude around earth = 24 hours time difference $\frac{360}{24}^\circ \text{ longitude} = 1 \text{ hours time difference}$ $15^\circ \text{ longitude} = 1 \text{ hours time difference}$ $30^\circ \text{ longitude} = 2 \text{ hours time difference}$ Since Braveheart is further West it is behind Hollandia Hollandia is 2 hours ahead of Braveheart	C
10	Time difference is $+3 + 8 = 11 \text{ hours}$ Moskow is ahead since it is +UTC. Hence it is 11 hours ahead of 6 pm Thursday 6 pm + 11 hrs = 5 am Friday	D

11	Time difference is $-3 - -10 = 7$ hours Rio is ahead since it has a higher UTC. Hence it is 7 hours ahead of 10 pm 1st July 10pm + 7hrs = 5 am 2nd July	C
12	Standard time difference = $10 + 8 = 18$ hours and Sydney is ahead So 18 hours before 9:00 pm or 21:00 is 03:00 or 3 am Standard time 3:00 am standard time is 4:00 am daylight saving time still on 7th August Local time is 4:00 am 7th August	C

MS-M2 Working with Time		Marks
Section II Solution		
Question 13		
(a)	7:40 to 8:00 is 0:20 8:00 to 10:27 is 2:27 So 7:40 to 10:27 is $0:20 + 2:27 = 2:47$ 2 hours and 47 minutes	1
(b)	$19:43 + 2:28 = 21:71 = 22:11$ $22:11 + 0:18 = 22:29$	2
Question 14		
(a)	First 3 shifts = $7:15 + 8:35 + 7:46 = 23:36$ "Fourth shift = $32:00 - 23:36 = 6:24$ " Fourth shift was 6 hours and 24 minutes	2
Question 15		
(a)	First train after 7:20 is 7:33 which arrives at City at 8:36	1
(b)	12 min before 8:45 is 8:33 so train arriving before this is 8:21 8:21 arrival leaves Woolsey at 8:04	2
(c)	If she caught the 7:50 from Weston she arrived at Seymour at 8:31 but that tram so she would need to catch the earlier tram from Harris St which is 7:24 This does not go to Seymour however, so she would need to change at Brereton, E to get on her previous tram to arrive at 8:31.	2
Question 16		
(a)	Greatest is between $UTC + 8$ and $UTC - 7$ which is $8 + 7 = 15$ hours	1
(b)	Sofia ($UTC 0$) = 8 pm Jenna $8\text{ pm} + 8 = 4\text{ am}$ Brittney $8\text{ pm} - 7 = 1\text{ pm}$ Pablo $8\text{ pm} - 4 = 4\text{ pm}$ Jenna is the only one who answers early in the morning at 4 am	2

(c)	Time difference is 15 hours and Jenna is ahead Back 15 hours from 2: 00 pm on Wed 21st July (15 hours = 12 hours + 3 hours) Back 12 hours is 2: 00 am on Wed 21st July Back another 3 hours is 11: 00 pm on Tue 20th July (OR 2: 00 pm on Wed 21st July = 14: 00 on Wed 21st July) Back 14 hours is midnight on Tue 20th July Back another 1 hour is 11: 00 pm on Tue 20th July	2
Question 17		
(a)	Difference is $165^\circ + 150^\circ = 315^\circ$ (since one is <i>E</i> and the other <i>W</i>) So time difference is $\frac{315}{15} = 21$ hours So Utopia is 21 hours ahead.	1
(b)	Utopia is 21 hours ahead, so go back 21 hours. (OR Go back 24 hours, then forward 3 hours) Back 24 hrs is 6: 30 am on Friday 14th Forward 3 hrs from that is 9: 30 am on Friday 14th	1
(c)	Local time on <i>FI</i> of leaving is 21 hours before 2: 00 p. m. on Tuesday August 18th So back 24 hrs and forward 3 hrs 5: 00 p. m. on Monday August 17th Time for trip = $\frac{4500}{250} = 18$ hours So arrival is 18 hours after 5: 00 p. m. on Monday August 17th Forward 12 hours is 5: 00 a. m. on Tuesday August 18th Forward another 6 hours is 11: 00 a. m. on Tuesday August 18th (So, she arrives 3 hours before the time she left on <i>UI</i> time)	2
Question 18		
(a)	Nairobi is $3 + 5 = 8$ hours ahead of Washington 8 hours before of 6 am Tue is 10 pm Mon.	1
(b)	Wed 3 pm plus 10 hour flight is 1 am on Thur Washington time 8 hours after 1 am Thur is 9 am Thur.	2